

Building Northwind Trading on aspgen

A complete, production-oriented walkthrough of every architecture tier, UI framework, and flag aspgen supports — built around one real, multi-context enterprise solution.

DOCUMENT	AUDIENCE	STATUS	THEME
Enterprise Developer Guide	.NET engineers adopting aspgen	v1.0, 2026-08-05	aspgen Document Theme

DECISION aspgen generates ASP.NET Core Clean Architecture APIs and Prism/Dryloc WPF apps from embedded Go `text/template` sources — nothing is downloaded at generation time and nothing runs as a service. The recommended entry point is the **context/arch engine** (`--context` / `--arch` / `-ui`), which lets one solution mix independent bounded contexts at four architecture tiers (`ar` , `dm` , `cqrs` , `es`) and attach one UI framework (`wpf` , `blazor` , `mvc` , or `spa`) on top. An older, still fully supported `--app` / `--backend` **workflow** (including the Renoir-style DDD Blazor profile) remains available for teams already using it. This guide builds a real multi-context system — **Northwind Trading** — end to end on the modern engine, then tours the legacy workflow and every flag aspgen accepts.

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1. What you'll build

Northwind Trading is a fictional wholesale distributor with four bounded contexts, each modeled at the architecture tier that actually fits its complexity — not the tier that looks the most impressive:

Context	Business need	Tier	Why this tier
Catalog	Public product lookup, high read volume, few business rules	<code>ar</code>	Flat CRUD is the right amount of ceremony for "here is a product record."
Inventory	Stock levels, reservations, warehouse transfers — invariants matter	<code>dm</code>	Needs an aggregate root protecting invariants, but no separate command/query host yet.
Sales	Order capture, pricing, fulfillment — multiple verbs, a real API	<code>cqrs</code>	Vertical-slice commands/queries plus a WebApi host justify the extra layer.

Context	Business need	Tier	Why this tier
	surface		
Billing	Invoices and ledger entries — full auditability, "what happened and when"	es	Event sourcing gives a true audit trail and time-travel debugging for money.

```

flowchart LR
    subgraph Solution["NorthwindTrading.sln"]
        Catalog["Catalog\n(ar)"]
        Inventory["Inventory\n(dm)"]
        Sales["Sales\n(cqrs)"]
        Billing["Billing\n(es)"]
    end
    Desktop["Desktop\nPrism/DryIoc WPF shell"]
    Storefront["External storefront\n(not generated)"]

    Storefront -->|REST| Catalog
    Desktop -->|in-process CrudService| Inventory
    Desktop -->|HTTP| Sales
    Desktop -->|HTTP| Billing

```

One **wpf** UI is attached to the solution, giving warehouse, sales, and finance staff a single Prism/DryIoc desktop shell with a module per context — because **wpf** is the only UI framework that spans **dm**, **cqrs**, and **es** tiers in one project. Catalog stays headless: **ar**-tier contexts get a real Minimal API host but no aspgen-generated UI screens, which is the correct trade-off for a lean, high-traffic read API meant to be called directly (curl, Postman, a separate storefront app, or a hand-written SPA).

Section 10 then tours the other three UI frameworks (**blazor** , **mvc** , **spa**) each in their own small, focused project, since a single project only gets one attached UI.

2. How aspgen thinks: two workflows, one generator

aspgen is a two-step generator, always:

1. `new` initializes a project once — `tree`, `.sln`, and `.aspgen/manifest.json`.
2. `add` (and `import-db`) mutate that already-generated project incrementally. `add` never creates a missing project; if `--project` is omitted it searches the current directory and its parents for `.aspgen/manifest.json`.

Everything is rendered from Go `text/template` sources embedded in the `aspgen` binary (`internal/templates/files/**`) — export and inspect them any time with `aspgen templates export`.

Two independent flag surfaces select the generation profile:

Workflow	Trigger flag	Status
Context/arch engine	<code>--context NAME --arch ar\ dm\ cqrs\ es</code>	Recommended for all new projects.
Legacy <code>--app / --backend</code>	<code>--app webapi\ wpf\ blazor\ fullstack</code>	Fully supported, no longer the advertised default.

The two are mutually exclusive per project — pick one when you run `new`. `--context` on the command line is what routes `new / add` into the newer engine; a bare `--app` (or no flags at all, which defaults to `--app webapi`) stays on the legacy path.

Callout — flag forms. Every `aspgen` flag accepts three equivalent forms: `--flag value`, `--flag:value`, and `-flag:value`. This guide uses the space form throughout; use whichever reads best in your own scripts.

3. Install and verify aspgen

```
git clone <this-repo>
cd aspgen
go build ./cmd/... ./internal/...
```

```
go run ./cmd/aspgen version
go run ./cmd/aspgen --help
```

`go run ./cmd/aspgen` is used for every command in this guide; substitute a built `aspgen.exe` if you prefer. aspgen never shells out to `dotnet` itself — you build and test the generated solution with your normal .NET tooling.

4. The architecture ladder — `ar` → `dm` → `cqrs` → `es`

The four tiers are an ordinal ladder: each is a strict superset of the concepts in the tier before it.

Tier	Adds on top of the previous tier	Host project	add kinds available
<code>ar</code>	Flat entity, direct <code>DbContext</code> CRUD	WebApi (Minimal API)	<code>add entity</code>
<code>dm</code>	Aggregate root, value objects, domain services, repository (interface + EF impl), event record scaffolds, synchronous <code>CrudService</code>	(none — <i>class libraries only</i>)	<code>add aggregate</code> , <code>add value-object</code> , <code>add domain-service</code> , <code>add repository</code> , <code>add event</code>
<code>cqrs</code>	Vertical-slice Application layer: Command/Query + Handler + FluentValidation validator per verb	WebApi (Minimal API)	all of <code>dm</code> 's kinds, <code>add repository</code> registers with DI
<code>es</code>	Append-only event store, event-sourced aggregates (<code>Raise</code> / <code>ApplyHistoric</code> / <code>LoadFromHistory</code>), synchronous read-model projections	WebApi (Minimal API)	all of <code>dm</code> 's kinds except <code>add repository</code> (already generated)

`ar` Active Record: flat entity, direct `DbContext` CRUD.

`dm` Domain Model: aggregate root + value objects + domain services + repository + ev

```
synchronous Application-layer CRUD service. No host project (class libraries on  
cQRS dm's Domain layer + vertical-slice Application (Command/Query/Handler/Validator  
verb) + a WebApi Minimal API host.  
es cQRS tier + an append-only event store, event-sourced aggregates (rehydrated by  
replaying events), and read-model projections.
```

A single solution may freely mix tiers across contexts — that's the point of the engine, and exactly how Northwind Trading is built below.

5. Scaffolding the solution shell

Bootstrap the solution with its first context, Catalog, at the `ar` tier:

```
go run ./cmd/aspgen new NorthwindTrading `
  --context Catalog --arch ar `
  --database sqlite `
  -ui wpf `
  --output ./NorthwindTrading
```

This single command:

- creates `NorthwindTrading.sln` plus the Catalog context's Domain/Features/WebApi tree at the `ar` tier;
- attaches the `wpf` UI (an empty Prism/Dryloc `Desktop` shell — Catalog is `ar`-tier, so it gets no screens yet; screens appear once `dm` / `cQRS` / `es` aggregates exist);
- writes `tests\NorthwindTrading.UnitTests` and `tests\NorthwindTrading.IntegrationTests` (every context with a WebApi host gets integration tests; pass `--no-tests` to skip both);
- writes `scripts\ci.ps1`, a local build/test/publish driver;
- writes `.aspgen/manifest.json`, the source of truth every later `add` command reads.

```
NorthwindTrading/
├─ NorthwindTrading.sln
├─ .aspgen/manifest.json
```

```

├─ scripts/
│   └─ ci.ps1
├─ src/
│   ├── NorthwindTrading.DomainModel/
│   │   └─ Catalog/                (ar entities live directly here)
│   ├── NorthwindTrading.Application/
│   ├── NorthwindTrading.Persistence/
│   ├── NorthwindTrading.Resources/
│   ├── WebApi/
│   │   ├── Program.cs
│   │   └─ Features/Catalog/
│   └─ Desktop/                    (Prism/DryIoc shell, from -ui wpf)
└─ tests/
    ├── NorthwindTrading.UnitTests/
    └─ NorthwindTrading.IntegrationTests/

```

The remaining three contexts are added with `add context` :

```

go run ./cmd/aspgen add context Inventory --arch dm --project ./NorthwindTrading
go run ./cmd/aspgen add context Sales --arch cqrs --project ./NorthwindTrading
go run ./cmd/aspgen add context Billing --arch es --project ./NorthwindTrading

```

6. Context 1 — Catalog (ar): a lean, headless product API

Catalog holds `Product` and `Category` , with a many-to-one relation from product to category:

```

go run ./cmd/aspgen add entity Category name:string --context Catalog --project ./NorthwindTrading
go run ./cmd/aspgen add entity Product name:string sku:string price:decimal active:boolean --context Catalog --project ./NorthwindTrading

```

`category:Category` synthesizes a required `CategoryId` foreign key and a `Category` navigation property (`nav:Entity?` would make it optional). Property types accepted by `name:type` are `string` , `int` , `long` , `decimal` , `float` , `bool` , `date` (`DateOnly`), `datetime` (`DateTime`), and `guid` / `uuid` ; append `?` for nullable (`middleName:string?`). Unknown types are a hard error, not a silent pass-through.

ar -tier endpoints are plain Minimal APIs against `DbContext` , including list, paged search, get-by-id, create, update, and delete:

```
// src/WebApi/Features/Catalog/Product/ProductEndpoints.cs (rendered)
public static class ProductEndpoints
{
    public static void MapProductEndpoints(this WebApplication app)
    {
        var group = app.MapGroup("/api/catalog/product");
        group.MapGet("/", async (AppDbContext db, CancellationToken cancellationToken)
            Results.Ok(await db.Products.AsNoTracking().ToListAsync(cancellationToken)
        group.MapGet("/search", async (string? search, decimal? priceMin, decimal? priceMax, decimal? active,
            int page, int pageSize, AppDbContext db, CancellationToken cancellationToken)
        {
            if (page < 1) page = 1;
            if (pageSize < 1) pageSize = 25;
            var query = db.Products.AsNoTracking().AsQueryable();
            if (!string.IsNullOrEmpty(search)) query = query.Where(x => x.Name.Contains(search));
            if (priceMin.HasValue) query = query.Where(x => x.Price >= priceMin.Value);
            if (priceMax.HasValue) query = query.Where(x => x.Price <= priceMax.Value);
            if (active.HasValue) query = query.Where(x => x.Active == active.Value);
            var totalCount = await query.LongCountAsync(cancellationToken);
            var items = await query.OrderBy(x => x.Id).Skip((page - 1) * pageSize).Take(pageSize).ToListAsync(cancellationToken);
            return Results.Ok(new { items, totalCount, page, pageSize, hasMore = (long)totalCount > (page * pageSize) });
        });
        group.MapGet("/{id:long}", async (long id, AppDbContext db, CancellationToken cancellationToken)
            Results.Ok(await db.Products.FindAsync(id, cancellationToken));
        group.MapPost("/", async (Product request, AppDbContext db, CancellationToken cancellationToken)
            Results.Ok(await db.Products.AddAsync(request, cancellationToken));
        group.MapPut("/{id:long}", async (long id, Product request, AppDbContext db, CancellationToken cancellationToken)
            Results.Ok(await db.Products.UpdateAsync(request, id, cancellationToken));
        group.MapDelete("/{id:long}", async (long id, AppDbContext db, CancellationToken cancellationToken)
            Results.Ok(await db.Products.RemoveAsync(id, cancellationToken));
    }
}
```

Because Catalog has no attached UI screens, this is a real, callable REST API today: run the WebApi host and hit `/api/catalog/product/search?search=widget&page=1&pageSize=25` , or point OpenAPI/Scalar at it (`/openapi/v1.json` , `/scalar/v1`).

Importing Catalog from an existing database

If Catalog already exists as a legacy database, skip hand-typed properties entirely:


```
go run ./cmd/aspgen import-db --project ./NorthwindTrading `
  --script schema.sql --provider postgres --tables Products,Categories --context Catal
```

`import-db` maps each selected table to one entity (best-effort singularized, PascalCased) through the same code path `add entity` uses. Primary keys and (on DDD-flavored backends) audit columns are skipped automatically; unmappable column types are skipped with a warning instead of failing the whole table. A `schema.sql` backup snapshot is written at the project root every run, and **no connection string is ever persisted** to the manifest, the backup, or any generated file — `dotnet ef migrations add / database update` stay a manual step you run yourself.

7. Context 2 — Inventory (`dm`): aggregates, value objects, domain services, repositories, events

`dm` is where the DDD building blocks appear: an aggregate root, immutable value objects, stateless domain services, an aggregate-scoped repository, and domain events — all as a synchronous, headless class-library graph (Domain ← Application ← Infrastructure/Persistence, no host until a UI attaches).

```
go run ./cmd/aspgen add aggregate StockItem sku:string quantityOnHand:int reorderPoint
go run ./cmd/aspgen add value-object BinLocation aisle:string shelf:string --context I
go run ./cmd/aspgen add domain-service ReplenishmentPolicy --context Inventory --proje
go run ./cmd/aspgen add repository StockItemRepository --aggregate StockItem --context
go run ./cmd/aspgen add event StockDepletedEvent stockItemId:long quantityOnHand:int -
```

The aggregate is a partial class split across two files — state/constructor and behavior — always generated together:

```
// src/NorthwindTrading.DomainModel/Inventory/StockItem.cs (rendered)
public sealed partial class StockItem : BaseEntity
{
    private StockItem() { }
```

```

public StockItem(string sku, int quantityOnHand, int reorderPoint)
{
    Sku = DomainGuard.Required(sku, nameof(sku));
    QuantityOnHand = quantityOnHand;
    ReorderPoint = reorderPoint;
}

public long Id { get; private set; }
public string Sku { get; private set; } = default!;
public int QuantityOnHand { get; private set; }
public int ReorderPoint { get; private set; }
// aspgen:navigation
}

```

```

// src/NorthwindTrading.DomainModel/Inventory/StockItem.Methods.cs (rendered)
public sealed partial class StockItem
{
    public void Update(string sku, int quantityOnHand, int reorderPoint)
    {
        Sku = DomainGuard.Required(sku, nameof(sku));
        QuantityOnHand = quantityOnHand;
        ReorderPoint = reorderPoint;
        Mark("system");
    }

    public override bool IsValid() => true;
    public override void Normalize() { }
    public override void Import(BaseEntity entity) { /* ... */ }
    public override string ToString() => $"StockItem #{Id}";
}

```

`BaseEntity` (shared across every `dm` / `cqrs` / `es` aggregate) provides soft delete and audit timestamps: `Deleted` , `Created` / `LastUpdated` `TimeStamp` values, and `Init(author)` / `Mark(author)` / `SoftDelete()` / `SoftUndelete()` . Mutating operations return `CommandResponse` (`Success` , optional `Message` / `Extra` , `Ok()` / `Fail()`) instead of a bare `bool` .

The generated `CrudService` is the synchronous Application-layer entry point every `dm` aggregate gets by default:

```
// src/NorthwindTrading.Application/StockItemCrudService.cs (rendered, trimmed)
public sealed class StockItemCrudService(NorthwindTradingDatabase database, IValidator<StockItemRequest> validator)
{
    public Task<List<StockItemView>> GetAllAsync(CancellationToken cancellationToken = default)
    {
        public async Task<StockItemView> CreateAsync(StockItemRequest request, CancellationToken cancellationToken)
        {
            await validator.ValidateAndThrowAsync(request, cancellationToken);
            var entity = new StockItem(request.Sku, request.QuantityOnHand, request.ReorderPoint);
            entity.Init("system");
            database.Add(entity);
            await database.SaveChangesAsync(cancellationToken);
            return new StockItemView(entity.Id, entity.Sku, entity.QuantityOnHand, entity.ReorderPoint);
        }
        public async Task<CommandResponse> UpdateAsync(long id, StockItemRequest request, CancellationToken cancellationToken)
        {
            public async Task<CommandResponse> DeleteAsync(long id, CancellationToken cancellationToken)
            {
                public async Task<(StockItemView[] Items, long TotalCount)> SearchAsync(string? sku, int? reorderPointMin, int? reorderPointMax, int page, int pageSize, CancellationToken cancellationToken)
                {
            }
        }
    }

    public sealed record StockItemRequest(string Sku, int QuantityOnHand, int ReorderPoint);
    public sealed record StockItemView(long Id, string Sku, int QuantityOnHand, int ReorderPoint);
}
```

Use `--no-crud` on `add aggregate` when a use case should be modeled with explicit domain behavior instead of starting from generated CRUD.

`add repository` generates and wires an aggregate-scoped repository contract (Domain) plus an EF implementation (Persistence), and **patches the CrudService to actually call it** rather than leaving it dead code: `CreateAsync` becomes `repository.AddAsync(entity, ct)`, `UpdateAsync` fetches via `repository.GetByIdAsync` and saves via `repository.SaveAsync`, and `DeleteAsync` collapses to a single `repository.DeleteAsync(id, ct)`. Reads (`GetAllAsync` / `SearchAsync`) deliberately keep querying the `DbContext` directly — a CQRS-style read/write split, not a shortcut.

```
src/NorthwindTrading.DomainModel/Inventory/
├─ StockItem.cs
├─ StockItem.Methods.cs
├─ BinLocation.cs (value object, immutable record)
├─ ReplenishmentPolicy.cs (domain service, stateless)
├─ IStockItemRepository.cs (repository contract)
└─ StockDepletedEvent.cs (domain event, immutable)
```

```
src/NorthwindTrading.Persistence/
├─ StockItemConfiguration.cs          (IEntityTypeConfiguration<StockItem>)
└─ Repositories/StockItemRepository.cs
src/NorthwindTrading.Application/
├─ StockItemCrudService.cs
└─ StockItemValidator.cs
```

Inventory has no host of its own — the WPF Desktop shell (added in Section 10) will call `StockItemCrudService` **in-process**, no HTTP involved.

8. Context 3 — Sales (`cqrs`): vertical-slice commands/queries and a WPF back office

`cqrs` keeps every `dm` building block and adds a WebApi host plus a genuine vertical-slice Application layer: one Command/Query, Handler, Validator per verb.

```
go run ./cmd/aspgen add aggregate Order number:string customer:string total:decimal p
go run ./cmd/aspgen add repository OrderRepository --aggregate Order --context Sales
```

```
src/NorthwindTrading.Application/Features/Sales/Order/
├─ CreateOrderCommand.cs
├─ CreateOrderHandler.cs
├─ UpdateOrderCommand.cs
├─ UpdateOrderHandler.cs
├─ DeleteOrderCommand.cs
├─ DeleteOrderHandler.cs
├─ GetOrderByIdQuery.cs
├─ GetOrderByIdHandler.cs
├─ GetOrdersQuery.cs
├─ GetOrdersHandler.cs
├─ SearchOrdersQuery.cs
├─ SearchOrdersHandler.cs
└─ OrderPagedResponse.cs
src/WebApi/Features/Sales/Order/
└─ OrderEndpoints.cs
```

Each Handler is a thin wrapper around the same `CrudService` the `dm` tier already generates — no duplicate repository contract is invented:

```
// Features/Sales/Order/CreateOrderCommand.cs (rendered)
namespace NorthwindTrading.Application.Features.Sales.Order;

public sealed record CreateOrderCommand(string Number, string Customer, decimal Total,
```

```
// Features/Sales/Order/CreateOrderHandler.cs (rendered)
namespace NorthwindTrading.Application.Features.Sales.Order;

public sealed class CreateOrderHandler(OrderCrudService service) : IHandler<CreateOrderCommand>
{
    public async Task<Result<OrderView>> HandleAsync(CreateOrderCommand request, CancellationToken token)
    {
        var view = await service.CreateAsync(new OrderRequest(request.Number, request.Customer, request.Total), token);
        return Result<OrderView>.Success(view);
    }
}
```

The WebApi host is a minimal `Program.cs` that mounts every feature via a marker comment kept up to date automatically:

```
// src/WebApi/Program.cs (rendered)
using NorthwindTrading.Application;
using NorthwindTrading.Infrastructure;

var builder = WebApplication.CreateBuilder(args);
builder.Services.AddApplication();
builder.Services.AddInfrastructure(builder.Configuration);

var app = builder.Build();
app.MapHealthChecks("/health");
// aspgen:features
app.MapGet("/", () => Results.Ok(new { application = "NorthwindTrading" }));
app.Run();

public partial class Program { }
```

Attach the WPF Desktop shell — already present from Section 5 — to Sales' aggregates by adding the aggregate first, then rendering (or refreshing) the UI:

```
go run ./cmd/aspngen add ui wpf --framework wpf --theme wpfui --theme-mode light --proj
```

`Order` 's generated `Store` calls the WebApi host over HTTP (Sales is `cqrs` -tier), while the `StockItem` store from Section 7 calls its `CrudService` in-process (Inventory is `dm` -tier) — both branches live in the same templated `{{.Name}}Store.cs.tpl`, selected by the aggregate's own tier:

```
// Desktop/Modules/Order/Services/OrderStore.cs (rendered, cqrs branch – HTTP)
public sealed class OrderStore(HttpClient http) : IOrderStore
{
    private const string Path = "/api/sales/order";
    public IReadOnlyList<OrderRow> GetAll() => http.GetFromJsonAsync<List<OrderRow>>(Path).GetAwaiter().GetResult();
    public OrderRow Save(long editingId, OrderRow value)
    {
        if (editingId == 0)
        {
            var response = http.PostAsJsonAsync(Path, value).GetAwaiter().GetResult();
            response.EnsureSuccessStatusCode();
            return response.Content.ReadFromJsonAsync<OrderRow>().GetAwaiter().GetResult();
        }
        http.PutAsJsonAsync($"{Path}/{editingId}", value).GetAwaiter().GetResult().EnsureSuccessStatusCode();
        return value with { Id = editingId };
    }
    // Delete(...), Search(...) follow the same HTTP pattern
}
```

```
// Desktop/Modules/StockItem/Services/StockItemStore.cs (rendered, dm branch – in-process)
public sealed class StockItemStore(StockItemCrudService service) : IStockItemStore
{
    public IReadOnlyList<StockItemRow> GetAll() =>
        service.GetAllAsync().GetAwaiter().GetResult()
            .Select(x => new StockItemRow(x.Id, x.Sku, x.QuantityOnHand, x.ReorderPoint))
            .ToList();

    public StockItemRow Save(long editingId, StockItemRow value)
```

```

{
    var request = new StockItemRequest(value.Sku, value.QuantityOnHand, value.ReorderPoint);
    if (editingId == 0)
    {
        var created = service.CreateAsync(request).GetAwaiter().GetResult();
        return new StockItemRow(created.Id, created.Sku, created.QuantityOnHand, created.ReorderPoint);
    }
    service.UpdateAsync(editingId, request).GetAwaiter().GetResult();
    return value with { Id = editingId };
}

public void Delete(long id) => service.DeleteAsync(id).GetAwaiter().GetResult();
}

```

`--theme wpfui` adds the WPF-UI Fluent theme (NuGet package, resource dictionaries, a themed `FluentWindow` shell); `--theme-mode light|dark` picks the starting palette (default `light`) — the shell also ships a runtime toggle, so this only affects the first paint.

9. Context 4 — Billing (`es`): event sourcing and an append-only ledger

`es` keeps everything `cqrs` has and rebuilds the aggregate around events instead of mutable state: an append-only event store with optimistic concurrency, an `EventSourcedAggregate` base class, and a synchronous read-model projection updated in the same unit of work as the event append.

```
go run ./cmd/aspgen add aggregate Invoice number:string customer:string amount:decimal
```

The aggregate replays its own history instead of storing current-state columns directly:

```

// src/NorthwindTrading.DomainModel/Billing/Invoice.cs (rendered, trimmed)
public sealed partial class Invoice : EventSourcedAggregate
{
    private Invoice() { }
}

```

```

public static Invoice LoadFromHistory(long id, IEnumerable<object> history)
{
    var aggregate = new Invoice { Id = id };
    foreach (var domainEvent in history) aggregate.ApplyHistoric(domainEvent);
    return aggregate;
}

public string Number { get; private set; } = default!;
public string Customer { get; private set; } = default!;
public decimal Amount { get; private set; } = default!;
public DateOnly IssuedOn { get; private set; } = default!;

protected override void Apply(object domainEvent)
{
    switch (domainEvent)
    {
        case InvoiceCreatedEvent created:
            Number = created.Number; Customer = created.Customer; Amount = created.Amount; IssuedOn = created.IssuedOn;
            break;
        case InvoiceUpdatedEvent updated:
            Number = updated.Number; Customer = updated.Customer; Amount = updated.Amount; IssuedOn = updated.IssuedOn;
            break;
        case InvoiceDeletedEvent:
            Deleted = true;
            break;
        // aspgen:events
        default:
            throw new DomainException($"Unhandled event {domainEvent.GetType().Name}");
    }
}
}

```

The event-store repository loads by replaying, saves by appending only the newly-raised events (optimistic concurrency via an expected-version check), and immediately projects the change into a queryable read model:

```

// src/NorthwindTrading.Application/InvoiceEventStoreRepository.cs (rendered, trimmed)
public sealed class InvoiceEventStoreRepository(EventStore eventStore, NorthwindTradingContext context)
{
    public async Task<Invoice?> LoadAsync(long id, CancellationToken cancellationToken)
    {

```



```

        var records = await eventStore.LoadAsync(nameof(Invoice), id, cancellationToken);
        return records.Count == 0 ? null : Invoice.LoadFromHistory(id, records.Select(r => r.Data));
    }

    public async Task SaveAsync(Invoice aggregate, CancellationToken cancellationToken)
    {
        var newEvents = aggregate.DequeueUncommittedEvents();
        if (newEvents.Count == 0) return;
        var expectedVersion = aggregate.Version - newEvents.Count;
        await eventStore.AppendAsync(nameof(Invoice), aggregate.Id, expectedVersion, newEvents, cancellationToken);
        await ProjectAsync(aggregate, cancellationToken);
    }

    private async Task ProjectAsync(Invoice aggregate, CancellationToken cancellationToken)
    {
        var readModel = await database.InvoiceReadModels.FirstOrDefaultAsync(x => x.Id == aggregate.Id);
        // add / update / remove InvoiceReadModel to match aggregate state, then Save
    }
}

```

`add repository` is rejected for `es`-tier aggregates — `{Aggregate}EventStoreRepository` already fills that role; there is no second contract to generate. Event versioning/schema evolution and snapshotting are deliberately out of scope for the generated code — plan for them yourself as the ledger grows.

```

src/NorthwindTrading.DomainModel/Billing/
├─ Invoice.cs
├─ Invoice.Methods.cs
└─ InvoiceEvents.cs (Created/Updated/Deleted event records)
src/NorthwindTrading.Application/
├─ InvoiceEventStoreRepository.cs
├─ InvoiceRequest.cs
└─ InvoiceValidator.cs
src/NorthwindTrading.Application/Features/Billing/Invoice/
├─ CreateInvoiceCommand.cs / Handler.cs
├─ UpdateInvoiceCommand.cs / Handler.cs
├─ DeleteInvoiceCommand.cs / Handler.cs
├─ GetInvoiceByIdQuery.cs / Handler.cs
├─ GetInvoicesQuery.cs / Handler.cs
└─ SearchInvoicesQuery.cs / Handler.cs

```

`Invoice` now shows up in the same Desktop shell alongside `Order` and `StockItem` — the WPF Store's `cqrs / es` branch is identical (both call the WebApi host over HTTP); only the WebApi handler underneath differs.

10. A tour of every UI framework

A UI attaches to the **whole project** — one UI framework surfaces every compatible aggregate in every context, which is why Northwind Trading above picked a single `wpf` shell spanning `dm` + `cqrs` + `es`. This section builds one small, focused project per remaining UI framework so each gets full coverage without overstating what one project can combine.

UI (-ui / --framework)	Compatible tiers	Transport	Notes
<code>wpf</code>	<code>dm</code> , <code>cqrs</code> , <code>es</code>	in-process (<code>dm</code>) or HTTP (<code>cqrs / es</code>)	Only UI spanning all three; supports <code>--theme wpfui / --theme-mode</code> .
<code>blazor</code>	<code>cqrs</code> , <code>es</code>	HTTP	Blazor Server calling the WebApi host.
<code>mvc</code>	<code>dm</code> only	in-process	Classic ASP.NET Core MVC, calls <code>CrudService</code> directly — no HTTP.
<code>spa</code>	<code>cqrs</code> , <code>es</code>	HTTP (bring your own frontend)	Wires OpenAPI/Scalar + permissive local-dev CORS onto the host; scaffolds no frontend project.

Every attached UI (`wpf` / `blazor` / `mvc`) renders a full list/edit/details CRUD screen for every already-present aggregate, and keeps generating a screen automatically for every aggregate `add` ed afterward.

`blazor` — a Sales web portal (`cqrs`)

```
go run ./cmd/aspgen new SalesPortal --context Sales --arch cqrs -ui blazor --output ./
go run ./cmd/aspgen add aggregate Order number:string customer:string total:decimal p
```

The Blazor host calls the WebApi over HTTP (not in-process — a deliberate difference from the legacy Renoir Blazor profile in Section 12), and each aggregate gets a Razor CRUD page with quick search and an advanced filter panel:

```
@* Components/Pages/Sales/OrderCrud.razor (rendered, trimmed) *@
@page "/sales/orders"
@inject NorthwindTrading.Application.OrderCrudService Service
<h1>Orders</h1>
<EditForm Model="form" OnValidSubmit="SaveAsync">
    <DataAnnotationsValidator />
    <div class="mb-3"><label>Number</label><InputText @bind-Value="form.Number" class=
    <div class="mb-3"><label>Customer</label><InputText @bind-Value="form.Customer" cl
    <div class="mb-3"><label>Total</label><InputNumber @bind-Value="form.Total" class=
    <button class="btn btn-primary" type="submit">Save</button>
</EditForm>
```

mvc — a warehouse admin dashboard (dm)

```
go run ./cmd/aspgen new WarehouseOps --context Inventory --arch dm -ui mvc --output ./
go run ./cmd/aspgen add aggregate StockItem sku:string quantityOnHand:int reorderPoint
```

MVC calls `CrudService` directly, in-process — no HTTP client, matching `dm`'s headless nature — with classic controller actions and query-string filters (no client-side state needed at all):

```
// Controllers/StockItemController.cs (rendered, trimmed)
[Route("inventory/stock-item")]
public sealed class StockItemController(StockItemCrudService service) : Controller
{
    private const int PageSize = 25;

    [HttpGet("")]
    public async Task<IActionResult> Index(string? search, int? quantityOnHandMin, int
    {
        if (page < 1) page = 1;
        var (items, totalCount) = await service.SearchAsync(search, quantityOnHandMin,
        return View(new StockItemIndexViewModel(items, totalCount, page, PageSize, se
    }
}
```

```
[HttpPost("create")]
public async Task<IActionResult> Create(StockItemRequest request, CancellationToken
{
    try { await service.CreateAsync(request, cancellation_token); return RedirectTo
    catch (ValidationException ex) { foreach (var f in ex.Errors) ModelState.AddMc
    }
    // Details, Edit, Delete follow the same pattern
}
```

spa — an API-first Billing service (es)

```
go run ./cmd/aspngen new BillingApi --context Billing --arch es -ui spa --output ./Bill
go run ./cmd/aspngen add aggregate Invoice number:string customer:string amount:decimal
```

spa scaffolds no frontend project at all — it patches `Program.cs / WebApi.csproj` in place with OpenAPI discovery, Scalar (`/scalar/v1`), and a permissive local-dev CORS policy, so any hand-written or separately generated frontend (React, Vue, Angular, a mobile app) can call the API immediately during development.

Attach any UI after the fact instead of at `new` time with `add ui` :

```
go run ./cmd/aspngen add ui spa --framework spa --project ./BillingApi
go run ./cmd/aspngen add ui wpf --framework wpf --project ./NorthwindTrading
go run ./cmd/aspngen add ui blazor --framework blazor --project ./SalesPortal
go run ./cmd/aspngen add ui mvc --framework mvc --project ./WarehouseOps
```

11. Extending every layer — recipes for real feature work

Domain layer

- **Add a property to an existing aggregate/entity** without hand-editing every generated layer:

```
powershell go run ./cmd/aspngen add entity-field StockItem binLocation:string --project
./NorthwindTrading
```

`entity-field` patches the already-rendered Domain class, persistence configuration, Application request/response records, Handlers/Validators/Endpoints, seed data, and every attached UI layer (WPF Row/View/ViewModel, Blazor form model, MVC view model) in place — no regeneration, no duplication.

- **Many-to-one relations:** `category:Category` (required) or `category:Category?` (optional) inside any `add entity` / `add aggregate` call, synthesizing the FK property and navigation.
- **Many-to-many relations:** `tags:Tag[]` synthesizes a join aggregate/entity (e.g. `ProductTag`) in the same context, with two required many-to-one relations back to both sides — no hand-written join table needed.

Application layer

- `cqrs` / `es` Handlers are intentionally thin — put orchestration logic in the Handler, not the Endpoint; keep the Endpoint a pure HTTP-shape adapter.
- `dm`'s `CrudService` is the natural place for synchronous cross-cutting business rules that don't need the vertical-slice ceremony yet — promote to `cqrs` when the API surface earns it.

Infrastructure / Persistence

- **Switch or add a database provider:**

```
powershell go run ./cmd/aspgen add database postgres --project ./NorthwindTrading
```

`sqlite` is the default everywhere; `postgres` is recorded in `.aspgen/manifest.json` and emitted into the EF Core provider registration, connection string, and DI setup. `dm`-tier contexts are class libraries with no host yet to attach a provider to — the provider takes effect once a UI or a `cqrs` / `es` host exists.

- **Import more tables later:** `import-db` is incremental too — rerun it with a fresh `--script` / `--connection` and `--tables` to add more entities/aggregates without touching what's already generated.

WebApi layer

- Feature endpoints for `cqrs / es` register themselves via the `// aspgen:features` marker in `Program.cs` — never hand-edit that block; let `add aggregate / add repository` own it.
- `/health`, `/openapi/v1.json`, and `/scalar/v1` are wired on every host tier that has one (`ar / cqrs / es`).

UI layer

- `add module NAME` adds a new Prism module shell to an existing WPF project (run `add ui wpf` first if the project has none yet) — legacy `--app / --backend` profile only.
- `add ui` is idempotent per framework: re-running it against a project that already has aggregates renders any screens that are still missing (e.g. after `add aggregate` added something new) without touching existing ones.

Tests and CI

Every context/arch project gets `tests\{Project}.UnitTests` (an in-memory `DbContext` smoke test) and, for any tier with a host (`ar / cqrs / es`), `tests\{Project}.IntegrationTests` (`WebApplicationFactory<Program>` hitting the root endpoint). `scripts\ci.ps1` drives restore/build/test and optionally publish:

```
.\scripts\ci.ps1           # restore, build, test
.\scripts\ci.ps1 -Publish  # ...and publish the WebApi host (skipped for headless c
.\scripts\ci.ps1 -SkipTests -Configuration Debug
```

12. The legacy `--app / --backend` workflow, including the Renoir DDD Blazor profile

The original workflow predates `--context / --arch` and remains fully supported. It's still the right choice if your team already standardized on it, or you specifically want the Renoir-style Blazor DDD profile (which has no equivalent in the newer engine).

Profile matrix

<code>webapi</code>	Clean Architecture Web API
<code>webapi --backend ddd</code>	Clean Architecture + DDD/CQRS CRUD
<code>webapi --simple</code>	Single-project Active Record-style CRUD API
<code>wpf</code>	Prism/DryIoc desktop shell and local UI modules
<code>wpf --backend ddd</code>	Local DDD + SQLite layers with no WebApi
<code>fullstack --backend ddd</code>	DDD/CQRS API + Prism/DryIoc WPF modules
<code>fullstack --simple</code>	Simple CRUD API + WPF modules connected by HttpClient
<code>blazor</code>	Renoir-style DDD Blazor profile (see below)

`--simple` and `--backend ddd` are mutually exclusive. `--theme wpfui` (plus `--theme-mode light|dark`) applies to any `wpf` / `fullstack` project. `--seed dummy [N]` (default 3 records per entity) works with `--simple` or `--backend ddd`, seeding at API/WPF startup for development environments only.

Worked example: RenoirCommerce (`--app blazor`)

```
go run ./cmd/aspgen new RenoirCommerce --app blazor --output ./RenoirCommerce
go run ./cmd/aspgen add context Catalog --project ./RenoirCommerce
go run ./cmd/aspgen add aggregate Product name:string price:decimal active:bool public:bool
go run ./cmd/aspgen add value-object ProductCode value:string --context Catalog --project ./RenoirCommerce
go run ./cmd/aspgen add domain-service PricingPolicy --context Catalog --project ./RenoirCommerce
go run ./cmd/aspgen add repository ProductRepository --aggregate Product --context Catalog --project ./RenoirCommerce
go run ./cmd/aspgen add event ProductPriceChanged productId:long price:decimal --context Catalog --project ./RenoirCommerce
```

The Renoir profile's structural conventions are modeled directly on a real reference implementation (the Renoir sample from Dino Esposito's *Clean Architecture with .NET*), kept async instead of the reference's synchronous style — the same `BaseEntity` / `CommandResponse` / partial-class-aggregate conventions used by the `dm` + engine tiers above actually originate here. The generated Blazor CRUD page renders in-process (no HTTP client, unlike the newer `blazor-context` UI in Section 10) and never binds directly to the domain aggregate — it uses a local editable form model plus the same immutable `Request` / `View` records:

```
@* Components/Pages/Catalog/ProductCrud.razor (rendered, trimmed) *@
@page "/catalog/products"
@inject NorthwindTrading.Application.ProductCrudService Service
```

```

<h1>Products</h1>
<EditForm Model="form" OnValidSubmit="SaveAsync">
    <DataAnnotationsValidator />
    <div class="mb-3"><label>Name</label><InputText @bind-Value="form.Name" class="form-control" /></div>
    <div class="mb-3"><label>Price</label><InputNumber @bind-Value="form.Price" class="form-control" /></div>
    <div class="mb-3"><label>Active</label><InputCheckbox @bind-Value="form.Active" class="form-control" /></div>
    <div class="mb-3"><label>Published</label><InputDate @bind-Value="form.Published" class="form-control" /></div>
    <button class="btn btn-primary" type="submit">Save</button>
</EditForm>
<div class="mb-3 d-flex gap-2">
    <input @bind="searchText" @bind:event="oninput" class="form-control" placeholder="Search" />
    <button class="btn btn-outline-primary" @onclick="SearchAsync">Search</button>
</div>

```

Controls are chosen by declared type: `string` → `InputText`, `numeric` → `InputNumber`, `bool` → `InputCheckbox`, `date / datetime` → `InputDate`. Use `--no-crud` on `add aggregate` when a use case should be modeled explicitly instead of starting from generated CRUD.

Worked example: a Rails-style simple CRUD API + WPF fullstack

```

go run ./cmd/aspgen new MyApp --app fullstack --simple --theme wpfui --theme-mode dark
go run ./cmd/aspgen add entity Customer name:string age:int active:bool --project ./MyApp
go run ./cmd/aspgen add module Customers --project ./MyApp
go run ./cmd/aspgen add database postgres --project ./MyApp
go run ./cmd/aspgen add service Email --project ./MyApp
go run ./cmd/aspgen add feature CreateCustomer name:string age:int active:bool --project ./MyApp

```

This is one project, one Web API (Active Record-style, no DDD/CQRS layer), and Prism/Dryloc WPF modules connected over HTTP (`ASPGENT_API_URL`, default `http://localhost:5000`). `add feature` is the base Clean Architecture profile's vertical-slice add (request/response records, FluentValidation validator, CQRS handler returning `Result<T>`, Minimal API endpoint) — distinct from `--simple`'s flat entity CRUD and from the newer engine's `cqrs`-tier feature generation, but structurally similar.

Legacy `import-db`


```
go run ./cmd/aspgen new MyApp --app webapi --simple --script schema.sql --provider postgres
go run ./cmd/aspgen import-db --project ./MyApp --connection "file:demo.db" --provider postgres
go run ./cmd/aspgen new RenoirCommerce --app blazor --script schema.sql --provider sqlserver
```

Supported providers: `sqlite`, `postgres`, `sqlserver`, `mysql`. `--connection` and `--script` are mutually exclusive and both require `--provider`. On the `blazor` profile, `--context` is required since tables become aggregates, not flat entities.

13. Custom templates

Every rendered file comes from an embedded Go `text/template` tree. Export, edit, and point aspgen at your own copy:

```
go run ./cmd/aspgen templates export ./my-templates
go run ./cmd/aspgen templates list
go run ./cmd/aspgen templates validate ./my-templates
go run ./cmd/aspgen new MyApp --app webapi --templates ./my-templates
```

`--templates PATH` is also accepted by `new` in the context/arch engine. Validate a customized tree before pointing real generation at it — `templates validate` parses every template without rendering, catching syntax errors early.

14. Testing, CI, and production readiness

Verification checklist before shipping a change built on aspgen-generated code (or before trusting a generated project in CI):

1. `go build` / `go vet` / `go test` the generator itself if you're customizing templates (`./cmd/...` `./internal/...`, never bare `./...` — example assets under `agent/skills/**` don't compile standalone).
2. `templates validate` any custom template directory.

3. `dotnet restore / build / test` the generated solution — `scripts\ci.ps1` does this in one call and mirrors what CI runs.
4. Run generation twice against the same output and confirm no unwanted duplication (idempotency is a design goal, not an afterthought).
5. Review project references match the intended dependency direction: Domain $\leftarrow$ Application $\leftarrow$ Infrastructure $\leftarrow$ WebApi (and Desktop $\rightarrow$ Infrastructure/Application/Domain for `wpf -- backend ddd`).
6. Confirm `.aspgen/manifest.json` accurately reflects every context/aggregate/entity/UI — future `add` commands trust it completely.
7. For production, review `appsettings.json` connection strings yourself; aspgen never writes production secrets and never runs `dotnet ef` — migrations stay a manual, reviewed step.

15. Full flag reference

`aspgen new` — **context/arch engine**

Flag	Values	Notes
<code>--context CTX</code>	any name	Presence routes into the engine instead of <code>--app / -- backend</code> .
<code>--arch TIER</code>	<code>ar</code> , <code>dm</code> , <code>cqrs</code> , <code>es</code>	All four tiers implemented.
<code>-ui UI</code>	<code>wpf</code> , <code>blazor</code> , <code>spa</code> , <code>mvc</code>	See Section 10 for tier compatibility; omit for a headless project.
<code>--database DB</code>	<code>sqlite</code> (default), <code>postgres</code>	<code>dm</code> -tier has no host to attach the provider to yet.
<code>--output PATH</code>	directory	Default: project name.

Flag	Values	Notes
<code>--templates</code> PATH	directory	Use a custom template set instead of the embedded one.
<code>--no-tests</code>	flag	Skip <code>{Project}.UnitTests</code> / <code>{Project}.IntegrationTests</code> .
<code>--dry-run</code>	flag	Print planned changes without writing files.
<code>--force</code>	flag	Overwrite existing files.

`aspgen new` — **legacy** `--app` / `--backend`

Flag	Values	Notes
<code>--app</code> TARGET	<code>webapi</code> (default), <code>wpf</code> , <code>blazor</code> , <code>fullstack</code>	
<code>--simple</code>	flag	Rails-style Active Record CRUD; not with <code>--backend</code> <code>ddd</code> .
<code>--backend</code> <code>ddd</code>	flag	Clean Architecture DDD/CQRS layers (<code>webapi</code> / <code>fullstack</code> , or <code>wpf</code> for local-only DDD).
<code>--database</code> DB	<code>sqlite</code> (default), <code>postgres</code>	<code>webapi</code> / <code>fullstack</code> / <code>wpf+ddd</code> only.
<code>--theme</code> <code>wpfui</code>	flag	WPF-UI Fluent theme (<code>wpf</code> / <code>fullstack</code> only).
<code>--theme-</code> <code>mode</code> MODE	<code>light</code> (default), <code>dark</code>	Requires <code>--theme</code> <code>wpfui</code> .
<code>--seed</code> <code>dummy</code> [N]	count, default 3	Requires <code>--simple</code> or <code>--backend</code> <code>ddd</code> .
<code>--script</code> PATH	file	SQL DDL to import entities/aggregates from; requires <code>-</code> <code>-provider</code> .

Flag	Values	Notes
<code>--provider P</code>	<code>sqlite</code> , <code>postgres</code> , <code>sqlserver</code> , <code>mysql</code>	Required with <code>--script</code> .
<code>--tables T</code>	<code>all</code> (default) or comma list	
<code>--context CTX</code>	name	Required with <code>--script</code> on the <code>blazor</code> profile (tables become aggregates).

aspngen add — kinds

Kind	Usage	Profile
<code>entity</code>	<code>add entity NAME prop:type... [-context CTX]</code>	Simple/legacy, or <code>ar</code> -tier engine context with <code>--context</code> .
<code>entity-field</code>	<code>add entity-field NAME prop:type...</code>	Adds properties to an existing entity/aggregate, any profile.
<code>module</code>	<code>add module NAME</code>	WPF Prism module; legacy <code>--app / --backend</code> , run <code>add ui</code> first.
<code>database</code>	<code>add database NAME</code>	Register/switch the persistence provider.
<code>service</code>	<code>add service NAME</code>	Application service; legacy non-simple <code>webapi / fullstack</code> .
<code>feature</code>	<code>add feature NAME prop:type...</code>	Web API vertical-slice CRUD feature; legacy Clean Architecture profile.
<code>ui</code>	<code>add ui --framework wpf\ blazor\ spa\ mvc</code>	Legacy: WPF onto a webapi project. Engine: attach to a <code>--context / --arch</code> project.
<code>context</code>	<code>add context NAME [--arch TIER]</code>	Blazor/Renoir profile without <code>--arch</code> ; engine context with <code>--arch</code> .

Kind	Usage	Profile
aggregate	add aggregate NAME prop:type... --context CTX [--no-crud]	Blazor/Renoir profile, or dm + engine context.
value-object	add value-object NAME prop:type... --context CTX	Blazor/Renoir profile, or dm + engine context.
domain-service	add domain-service NAME -- context CTX	Blazor/Renoir profile, or dm + engine context.
repository	add repository NAME --aggregate AGG --context CTX	Blazor/Renoir profile, or dm + engine context (not es).
event	add event NAME prop:type... -- context CTX	Blazor/Renoir profile, or dm + engine context.

Common add flags: --project PATH (default: search up from cwd for .aspgen/manifest.json), --theme wpfui / --theme-mode MODE (for ui / module), --dry-run , --force .

aspgen import-db

Flag	Values	Notes
--project PATH	directory	Default: search up from cwd.
--script PATH	file	SQL DDL to parse (mutually exclusive with --connection).
--connection STR	connection string	Live introspection (mutually exclusive with --script).
--provider P	sqlite , postgres , sqlserver , mysql	Required.
--tables T	all (default) or comma list	
--backend ddd	flag	Override the project's own backend profile.

Flag	Values	Notes
<code>--context CTX</code>	name	Required for the Blazor/Renoir profile (tables → aggregates).
<code>--dry-run / --force</code>	flag	Same semantics as <code>new / add</code> .

Property type syntax

`name:type` pairs, nullable with a trailing `?` (e.g. `middleName:string?`). Accepted types: `string` , `int` , `long` , `decimal` , `float` , `bool` , `date` (→ `DateOnly`), `datetime` (→ `DateTime`), `guid` / `uuid` (→ `Guid`). Relations: `nav:Entity` (required many-to-one), `nav:Entity?` (optional many-to-one), `nav:Entity[]` (many-to-many, synthesizes a join entity/aggregate). Unknown types are always a hard error.

16. Troubleshooting and known gotchas

- `add` fails with "no manifest found" — you're outside the generated project tree and its parents; pass `--project PATH` explicitly.
- `spa / blazor / mvc` UI rejected on a context — check the tier compatibility table in Section 10; `spa / blazor` need `cqrs / es` , `mvc` needs `dm` , and `ar` -tier contexts get no generated UI screens at all by design.
- `add repository` rejected on an `es` aggregate — expected; the aggregate already has a generated `{Aggregate}EventStoreRepository` .
- `--theme-mode` rejected without `--theme wpfui` — the flag only has meaning alongside the WPF-UI Fluent theme.
- **Property type rejected** — only the type list in Section 15's property syntax table is supported; anything else is a deliberate hard error rather than a silent pass-through.
- **A generated entity is named something like `Task`** — avoid names that collide with common .NET/BCL types (`System.Threading.Tasks.Task` is the classic one); it compiles as ambiguous C#. Use a more specific domain name instead (`TodoItem` , not `Task`).

- `go test -race` on generated Go tooling changes fails locally with "requires cgo" — a local-machine toolchain gap, not a generator bug; CI runners have cgo enabled.
 - `dotnet restore` fails with NU1301 against a corporate NuGet feed — restore explicitly from `nuget.org` (`dotnet restore <project> -s https://api.nuget.org/v3/index.json`), then `dotnet build` normally.
-

17. Next steps

- Grow Northwind Trading incrementally: add `add aggregate ShipmentRoute ...` to Inventory, `add feature ApplyDiscount ...` under Sales, or a new `Returns` context at whichever tier its complexity actually earns.
 - Point `templates export` at a copy of the embedded set and adapt naming/status-code/validation conventions to your team's own standards before scaling this to a second real product.
 - Wire `scripts\ci.ps1` into your existing CI pipeline (`-Publish` on merge to main, plain restore/build/test on pull requests) so every generated context stays honestly buildable, not just generatable.
-